An Introduction to Module theory

A Study of Modules Over Rings

Course Information

Modern Algebra II

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This project is an exploration of Module Theory. The paper assumes decent knowledge of Linear Algebra, Group theory, Field and Ring theory including: irreducibility, principle ideal domains, ring homomorphisms, and isomorphism theorems. We aim to introduce concepts in Module Theory and investigate some but not all of their implications in Linear Algebra, providing a way to generalize the notion of vector spaces and linear transformations.

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1 Introduction

This project aims to explore basic concepts and theorems in Module Theory. Developed as a part of Emmy Noether work on Abstract Algebra, Module Theory is an extension of ring theory. In fact, it represents a bridge between Ring & Field theory and Linear Algebra as we shall see in this project. The first mathematician who introduced the term "Module" was Dedekind back in 1871. The word "Module" originates back to the German word "Modul". Dedekind defined a Module in his paper Vorlesungen über Zahlentheorie (Lectures on Number Theory) to be a system a (the old terminology for a set) of numbers closed under addition and subtraction (this is exactly a $\mathbb{Z}-$module). He also introduced the divisibility relation using the mod function and this was the reason behind why he called this system a module. He stated that if the difference of two numbers belongs to the system then they are congruent modulo a written as $c \equiv b \mod (a)$. Dedekind went on to define the notion of an ideal of a number field. That is a set of algebraic numbers (introduced in the same way as algebraicity we know today) in a number field that are closed under addition, subtraction and multiplication by algebraic elements in the number field. He called ideals generated by a single element *principle ideals*. Later on, module theory has been developed by Emmy Noether. She first introduced the ascending chain condition, later proving that ideals in rings that satisfy such a condition are finitely generated. She also developed what now known as the isomorphism theorems of rings. All of these results preceded the modern theory we are introducing in this paper. A module in the modern sense is an abelian group that is being acted on by a ring. This definition allows us to say a module is simply a generalization of a vector space. In this project we mainly aim to investigate the structure theorem, a major result in module theory generalizing the fundamental theorem of finite abelian groups (which we will see later are just $\mathbb{Z}-$modules) to finitely generated modules over PIDs. This provides a useful tool to understand the structure of a module form the structure of the ring's ideals. In the second part of the paper, we introduce the rational canonical form of a linear transformation of a vector space. This result is motivated by modules over the ring of polynomials $F[x]$. In the next section, we introduce the basics of Module Theory.

2 Module Theory

2.1 Introduction

Module theory is an attempt to generalize the concept of vector spaces, as we will see later that vector spaces are just modules over fields, however, modules can be defined over any ring, not even necessarily over commutative rings, or rings with unity. Later we will see how we can use this powerful generalization to prove statements about ideals and quotient rings just by treating them as modules.

Definition 2.0: A Module over a Ring R

Let R be any ring, we call M a *left R-module* if the M is a set paired with the following

- a binary operation $+$, such that M is an Abelian group under the operation.
- an action of R on M, that is a map $R \times M \longrightarrow M$, denoted by rm, which satisfies the following for all $r, s \in R$, and for all $m, n \in M$:
  - a) $(r+s)m = mr + ms$ *(Distributivity of ring addition)*
  - b) $(rs)m = r(sm)$ *(Compatibility)*
  - c) $r(m+n) = rm + rn$ *(Distributivity over module addition)*

  If the ring R has a unity 1, we then add another axiom for the actions, $1m = m$, for all $m \in M$ and it's called unitality.

We call it *left* $R-$module because the elements of R always appear on the left, that actions of R on M, always look like rm, we can define right modules analogously, it then becomes clear that if R is commutative, then every left module can be made into a right module by defining the the action map of the right module to be $mr = rm$.

Definition 2.1: Submodule

Let R be a ring, and let M be an R-module, then a submodule N of M, is a subgroup of M, that is closed under the ring action, i.e, $rn \in N$ for every $n \in N$, $r \in R$.

Theorem 2.2: The Submodule Criterion

Let R be a ring and let M be an R-module. A subset N of M is a submodule of M if and only if

1. $N \neq \emptyset$

   and

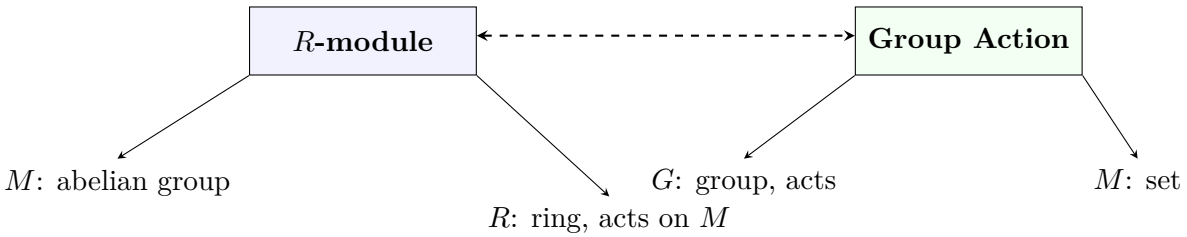
2. $x + ry \in N$ for all $r \in R$ and for all $x, y \in N$

Note 2.3

1. Note that in the following sections we assume that R is a commutative ring unless stated otherwise
2. We also may denote the action of R on a module M with rm for any $m \in M$ and $r \in R$ or $r \cdot m$

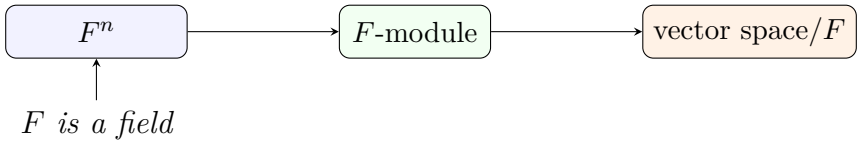
R-Modules

The central idea behind an R-module mirrors the notion of a group action: just as a group G acts on a set M, a ring R acts on an abelian group M.



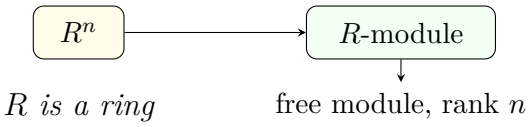
Example 2.4: Affine n-space over a field F

An affine n-space over a field F is F^n. Since F is a field, F^n is an abelian group under addition on which F acts by scalar multiplication, making F^n an F-module — that is, a *vector space* over F.



Example 2.5: R^n as the free module of rank n

If R is a ring, then R^n with componentwise addition and left multiplication by R is an R-module called the *free module over R of rank n*.



Remark 2.6

For a fixed ring R, different R-modules can be built on the *same* underlying abelian group M by varying the action of R on M. The distinguishing data is precisely *how* R acts.

Example 2.7

Consider the abelian group $M = \mathbb{Z}_6$ as a module over itself. Define the two actions:

1. $\varphi : M \to M$ such that $\varphi(m) = mn \mod 6$ for any $m, n \in \mathbb{Z}_6$
2. the trivial action $\phi(m) = 0$

In this case we define two different actions for the same abelian group and the same ring. We shall see more clear examples when we define $F[x]$-modules.

2.2 Annihilation by an Ideal

Definition 2.8: Annihilation by an ideal

Let M be an R-module and $I \trianglelefteq R$ an ideal. We say I *annihilates* M (equivalently, M is *annihilated by* I) if

$$am = 0 \qquad \forall\, a \in I, \quad \forall\, m \in M.$$

When I annihilates M, the R-module structure on M descends to an R/I-module structure via the action

$$(r + I)m \;=\; rm.$$

This is well-defined precisely *because* $am = 0$ for all $a \in I$: replacing r by $r + a$ does not change the action on m.

Remark 2.9

When I is a *maximal* ideal of R, the quotient R/I is a field $\mathbb{F}$, and M becomes a *vector space* over $\mathbb{F} = R/I$.

2.3 Abelian Groups as $\mathbb{Z}$-Modules

Example 2.10: Every abelian group is a $\mathbb{Z}$-module

Every abelian group M is naturally a $\mathbb{Z}$-module: the action of $n \in \mathbb{Z}$ on $m \in M$ is

$$nm \;=\; \underbrace{m + m + \cdots + m}_{n \text{ times}}.$$

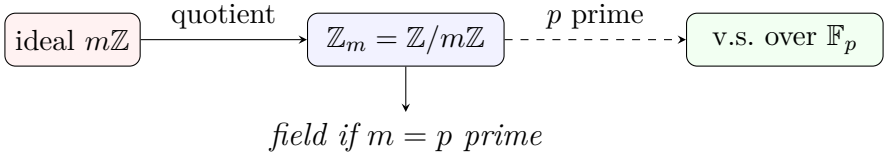
Example 2.11: Finite abelian groups are $\mathbb{Z}_m$-modules

Let A be an abelian group of finite order m. Then A is a module over $\mathbb{Z}_m = \mathbb{Z}/m\mathbb{Z}$, not merely over $\mathbb{Z}$.

Why? Because $mx = 0$ for every $x \in A$, the ideal $m\mathbb{Z}$ annihilates A:

$$ax = 0 \qquad \forall\, a \in m\mathbb{Z}, \quad \forall\, x \in A.$$

Hence $m\mathbb{Z} = \mathrm{Ann}(A)$ and the $\mathbb{Z}$-action descends to $\mathbb{Z}/m\mathbb{Z}$.



Module Homomorphisms

Definition 2.12: R-module homomorphism

Let M and N be R-modules. A map $\varphi : M \to N$ is an *R-module homomorphism* if for all $m_1, m_2 \in M$ and $r \in R$:

(i) $\varphi(m_1 + m_2) = \varphi(m_1) + \varphi(m_2)$,

(ii) $\varphi(rm) = r\,\varphi(m)$.

Remark 2.13

1. Not every group homomorphism between R-modules is an R-module homomorphism: the scalar-compatibility condition (ii) may fail.
2. All the isomorphism theorems for groups also hold for R-modules

Example 2.14: A group hom that need not be an R-module hom

Consider $\varphi : \mathbb{C} \to \mathbb{C}$, where $\mathbb{C}$ is a module over itself (the action of $\mathbb{C}$ is denoted by $\cdot$). Given by $\varphi(z) = \bar{z}$ where $\bar{z}$ is the conjugate of z. This function is a group homomorphism since $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$. In this case, $\varphi(i) = -i$, and $\varphi(1 \cdot i) = -i$ while $\varphi(1) \cdot i = i$. The condition $\varphi(rm) = r\varphi(m)$ fails, so φ need not be an R-module homomorphism even when it is a group homomorphism.

Theorem 2.15: Quotient Modules

Let R be a ring, let M be an R-module, and let N be a submodule of M. The additive abelian quotient group M/N can be made into an R-module by defining

$$r(x + N) = (rx) + N,$$

for all $r \in R$ and $x + N \in M/N$.

The natural projection map

$$\pi : M \to M/N$$

defined by

$$\pi(x) = x + N$$

is an R-module homomorphism with kernel N.

Remark 2.16

Note that a submodule is a normal subgroup of the ambient module. Hence, a quotient module is also an abelian quotient group.

3 Generation of Modules

Definition 3.0: Sum of submodules

Let $N_1, N_2, \ldots, N_n$ be submodules of M. Their *sum* is

$$N_1 + N_2 + \cdots + N_n \;=\; \Big\{ a_1 + a_2 + \cdots + a_n \;\Big|\; a_i \in N_i \;\forall\, i \Big\}.$$

Definition 3.1: Submodule generated by a set

Let $A \subseteq M$. The *submodule of M generated by A* is

$$RA \;=\; \Big\{ r_1 a_1 + r_2 a_2 + \cdots + r_k a_k \;\Big|\; a_i \in A,\; r_i \in R \Big\}.$$

If N is a submodule with $N = RA$ for some $A \subseteq M$, then N is *generated by* A and A is a *set of generators* of N (which need not be finite).

If $A = \{a\}$, we define the submodule generated by a with Ra where $Ra := RA$.

Definition 3.2: Basis

A subset $A = \{a_0, a_1, a_2, \ldots, a_n\}$ of an $R-$module M is a basis for M with respect to R if A generates M and for every $x \in M$ there exist unique $r_0, r_1, r_2, \ldots, r_n \in R$ such that $x = r_0 a_0 + r_1 a_1 + r_2 a_2 + \ldots r_n a_n$.

Exercise

Use the submodule criterion to verify that RA is a submodule of M for any $A \subseteq M$.

Definition 3.3: Finitely generated and cyclic submodules

- N is *finitely generated* if there exists a finite $A \subseteq M$ with $N = RA$.
- N is *cyclic* if there exists $a \in M$ with $N = Ra$.

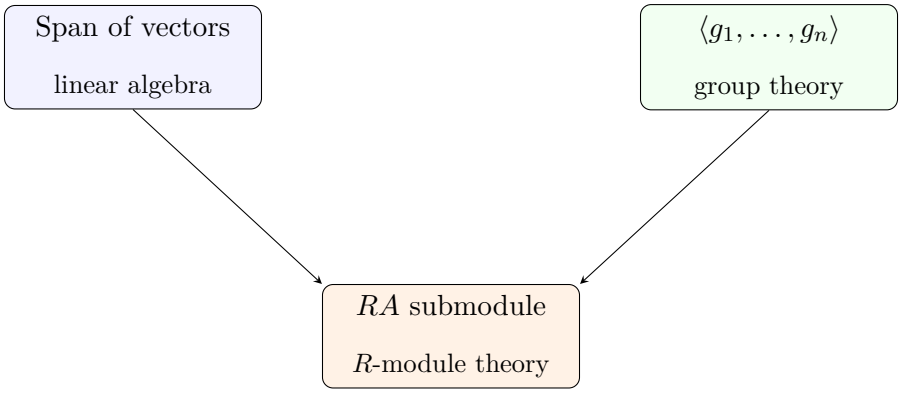
Remark 3.4

The submodule RA is the smallest submodule of M containing A analogous to the span of a set of vectors in linear algebra. In particular, the sum

$$N_1 + N_2 + \cdots + N_n$$

is the submodule generated by $N_1 \cup N_2 \cup \cdots \cup N_n$

If N is finitely generated it possesses a *minimal* generating set, echoing the notion of a basis as the minimal spanning set. This construction is analogous to two familiar ones:



Remark 3.5

How do we see $F[x]/\langle p(x)\rangle$ as a cyclic module? Consider M a cyclic $F[x]-$module generated by a non-zero element $m \in M$ i.e. every $n \in M$ is $n = f(x)m$ for some $f(x) \in F[x]$. Consider the surjective map $\phi : F[x] \to M$ given by $\phi(f(x)) = mf(x)$. By the first isomorphism theorem of rings we get that $F[x]/\ker\phi \cong M$. The $\ker(\phi) = \{f(x) \mid f(x)m = 0\}$, that is the kernel is the annihilator of m. Since m is the generator of M, then the kernel annihilates the whole module M i.e. $\mathrm{Ann}(M) = \ker(\phi)$. Since $F[x]$ is a PID and $\ker(\phi)$ is an ideal of $F[x]$. Then $\ker(\phi) = \langle p(x)\rangle$, where $p(x)$ is an annihilator of M of least degree.

4 Torsion Modules

Definition 4.0: Torsion element and torsion submodule

Let M be an R-module. An element $m \in M$ is a *torsion element* if $rm = 0$ for some nonzero $r \in R$.

When R is an *integral domain*, the set

$$\mathrm{Tor}(M) \;=\; \{\, m \in M \mid rm = 0 \text{ for some nonzero } r \in R \,\}$$

is a submodule of M called the *torsion submodule*.

Definition 4.1: Torsion module

An R-module M is a *torsion module* if every element is a torsion element:

$$\forall\, m \in M \quad \exists\, r \in R,\ r \neq 0, \quad rm = 0.$$

Note: the annihilating r may differ from element to element.

Note 4.2

Let R be a ring and view R as an R-module over itself.

1. The $\mathrm{Tor}(R)$ is *not* a submodule in general.

   Let R be any ring, and M be an $R-$module. Let $\mathrm{Tor}(M)$ be the set of all torsion elements in M. Pick any two elements m, n in $\mathrm{Tor}(M)$, where $rm = 0$ and $sn = 0$ for some $r, s \in R$. Therefore to check if a nonzero element in R annihilates the sum. Consider the following equation $rs(m+n) = 0$, where r, s are non-zero. If R is not an integral domain it is not necessarily the case that $rs \neq 0$.

   Counterexample: Consider the module $\mathbb{Z}_6$, where

   - $3 \times 2 = 0$
   - $2 \times 3 = 0$

   So, both $2, 3$ are torsion elements However the element $2+3 = 6$ is not annihilated by any of the elements of the ring $\mathbb{Z}_6$. Then the $\mathrm{Tor}(M)$ is not a submodule since it is not closed under addition.

2. Every finite abelian group is a torsion $\mathbb{Z}$-module.

   Proof. Let $\mathbb{Z}_m$ be a finite abelian group. Since the order of the group m is finite, then by lagrange theorem the order of every element g in $\mathbb{Z}_m$ is a divisor of m in particular the order is non-zero. Thus, $\underbrace{g + g \cdots + g}_{|g| \text{ times}} = 0$. Therefore every element g is annihilated by the integer $|g|$, and so every element is a torsion element. □

3. An example of an *infinite* abelian group that is a torsion $\mathbb{Z}$-module is the polynomial ring (hence abelian group under addition) $\mathbb{Z}_p[x]$, where every polynomial when added to itself $p-$times we get the zero polynomial. In general, any infinite ring of non-zero characteristic is a torsion $\mathbb{Z}-$module.

Theorem 4.3: Finitely generated torsion modules have a nonzero annihilator

Let R be an integral domain. Every finitely generated torsion R-module M has a *nonzero annihilator*: there exists a nonzero $r \in R$ such that

$$rm = 0 \qquad \forall\, m \in M.$$

(The *same* r works for all elements simultaneously, though there may be more than one such r.)

Proof. Let M be a finitely generated $R-$module. Since M is finitely generated, there exist a subset $A = \{a_1, a_2, \ldots, a_n\} \subseteq M$ such that

$$M = \{r_1a_1 + r_2a_2 + \ldots r_na_n \mid r_i \in R, a_i \in A\}$$

Thus every element $m \in M$ can be written as

$$m = r_1a_1 + r_2a_2 + \cdots + r_na_n$$

for some $r_1, r_2, \ldots, r_n \in R$.

Since M is a torsion module, for each a_i there exists a nonzero element $s_i \in R$ such that

$$s_ia_i = 0. \quad \text{for all } (1 \leq i \leq n)$$

Let

$$\alpha = s_1s_2 \cdots s_n.$$

Then $\alpha \neq 0$ since R is an integral domain.

Now for any $m \in M$,

$$\begin{aligned} \alpha m &= \alpha(r_1a_1 + r_2a_2 + \cdots + r_na_n) \\ &= \alpha r_1a_1 + \alpha r_2a_2 + \cdots + \alpha r_na_n. \end{aligned}$$

Since α is a multiple of each s_i , we have

$$\alpha a_i = 0$$

for every i. Since the action of R is commutative, then

$$\begin{aligned} \alpha m &= \alpha a_1 r_1 + \alpha a_2 r_2 + \cdots + \alpha a_n r_n \\ &= 0 + 0 + \cdots + 0 = 0. \end{aligned}$$

Therefore α annihilates every element of M, so M has a non-zero annihilator. □

Definition 4.4: Annihilator of a module

Let M be an $R-$module. We define the *annihilator* of M to be the following set

$$\mathrm{Ann}(M) \;=\; \{\, r \in R \mid rm = 0 \;\; \forall\, m \in M \,\}.$$

5 Direct Sums

Definition 5.0: Direct product and external direct sum

Let $M_1, M_2, \ldots, M_k$ be R-modules. Their *direct product* is

$$M_1 \times M_2 \times \cdots \times M_k \;=\; \{(m_1, m_2, \ldots, m_k) \mid m_i \in M_i\},$$

with componentwise addition and R-action. This is also written as the *external direct sum* $M_1 \oplus M_2 \oplus \cdots \oplus M_k$.

Theorem 5.1: Characterisation of internal direct sums

Let $N_1, N_2, \ldots, N_k$ be submodules of an R-module M. The following three conditions are equivalent:

(1) The map $\pi : N_1 \times N_2 \times \cdots \times N_k \longrightarrow N_1 + N_2 + \cdots + N_k$ given by

$$\pi(a_1, a_2, \ldots, a_k) = a_1 + a_2 + \cdots + a_k$$

is an isomorphism.

(2) $N_j \cap \left(\sum_{i \neq j} N_i\right) = \{0\}$ for every $j \in \{1, \ldots, k\}$.

(3) Every $x \in N_1 + N_2 + \cdots + N_k$ can be expressed *uniquely* as $x = a_1 + a_2 + \cdots + a_k$ with $a_i \in N_i$.

Proof. (1) $\Rightarrow$ (2)

Let

$$\phi : N_1 \times N_2 \times \cdots \times N_k \to N_1 + N_2 + \cdots + N_k$$

be the natural homomorphism defined by

$$\phi(n_1, n_2, \ldots, n_k) = n_1 + n_2 + \cdots + n_k.$$

Suppose for contradiction that there exists some j such that

$$S = N_j \cap \sum_{i \neq j} N_i \neq \{0\}.$$

Pick any nonzero element $n \in S$. Then $n = n_j$ for some $n_j \in N_j$, and also

$$n = n_1 + n_2 + \cdots + n_k$$

where each $n_i \in N_i$ and not all are zero for $i \neq j$.

Hence

$$\phi(n_j, 0, 0, \ldots, 0) = n_j$$

and

$$\phi(n_1, n_2, \ldots, n_k) = n_1 + n_2 + \cdots + n_k = n_j.$$

This contradicts injectivity of ϕ, since

$$(n_j, 0, 0, \ldots, 0) \neq (n_1, n_2, \ldots, n_k)$$

in the domain of ϕ.

Therefore $N_j \cap \sum_{i \neq j} N_i = \{0\}$ for every j, proving the intersection is trivial.
(2) $\Rightarrow$ (3) Suppose

$$N_j \cap \sum_{i \neq j} N_i = \{0\}$$

for every $j \in \{1, \ldots, k\}$. Suppose for contradiction that some $c \in N_1 + N_2 + \cdots + N_k$ has two different representations such that $c = a_1 + a_2 + \ldots a_k$ and $c = b_1 + b_2 + \cdots + b_k$, for

some $a_i, b_i \in N_i$ where

$$a_1 + a_2 + \cdots + a_k = b_1 + b_2 + \cdots + b_k$$

and $a_i \neq b_j$ for some i, j. WLOG suppose $a_1 \neq b_1$. Hence

$$a_1 - b_1 = b_2 - a_2 + b_3 - a_3 + \cdots + b_k - a_k \neq 0.$$

Whence $a_1 - b_1 \in N_i$, since N_i is a submodule and $a_1, b_1 \in N_i$, and $a_1 - b_1 \in \sum_{j=2}^{k} N_j$. Thus $a_1 - b_1$ is a non-zero element in the intersection $N_1 \cap \sum_{j=2}^{k} N_j$, contradicting our assumption. Thus c must have a unique representation as desired.

(3) $\Rightarrow$ (1)

Suppose that every element

$$x \in \sum_{i=1}^{k} N_i$$

is uniquely expressed as

$$x = a_1 + a_2 + \cdots + a_k,$$

where $a_i \in N_i$.

Define

$$\phi : N_1 \oplus N_2 \oplus \cdots \oplus N_k \to N_1 + N_2 + \cdots + N_k$$

given by

$$\phi(a_1, a_2, \ldots, a_k) = a_1 + a_2 + \cdots + a_k.$$

It is clear that ϕ is surjective.

To show injectivity, suppose

$$\phi(a_1, a_2, \ldots, a_k) = \phi(b_1, b_2, \ldots, b_k).$$

Then by definition of ϕ

$$a_1 + a_2 + \cdots + a_k = b_1 + b_2 + \cdots + b_k.$$

and by uniqueness of representation, we get that $a_i = b_i$ for every i. Hence

$$(a_1, a_2, \ldots, a_k) = (b_1, b_2, \ldots, b_k),$$

so ϕ is injective.

Therefore ϕ is a bijection.

Finally, we will show that ϕ is an R-module homomorphism. Clearly ϕ preserves addition. Consider $(a_1, a_2, \ldots, a_k), (b_1, b_2, \ldots, b_k) \in N_1 \times N_2 \times \cdots \times N_k$. Then

$$\begin{aligned}
\phi\Big((a_1, a_2, \ldots, a_k) + (b_1, b_2, \ldots, b_k)\Big) &= \phi(a_1 + b_1, a_2 + b_2, \ldots, a_k + b_k) \\
&= (a_1 + b_1) + (a_2 + b_2) + \cdots + (a_k + b_k) \\
&= (a_1 + a_2 + \cdots + a_k) + (b_1 + b_2 + \cdots + b_k) \\
&= \phi(a_1, a_2, \ldots, a_k) + \phi(b_1, b_2, \ldots, b_k).
\end{aligned}$$

To prove preserving the action of R, let $r \in R$. Then by axioms of modules

$$\begin{aligned}
\phi\Big(r(a_1, a_2, \ldots, a_k)\Big) &= \phi(ra_1, ra_2, \ldots, ra_k) \\
&= ra_1 + ra_2 + \cdots + ra_k \\
&= r(a_1 + a_2 + \cdots + a_k) \\
&= r\,\phi(a_1, a_2, \ldots, a_k).
\end{aligned}$$

Thus ϕ is an R-module isomorphism. □

Note 5.2

All three conditions concern the addition in M and the intersections of submodules. They capture the idea that the summands "do not overlap."

If $M = N_1 + N_2 + \cdots + N_k$ satisfies these conditions, M is the *internal direct sum* of $N_1, \ldots, N_k$, written

$$M = N_1 \oplus N_2 \oplus \cdots \oplus N_k.$$

Since $M \cong N_1 \times N_2 \times \cdots \times N_k$, the notations $\oplus$ and $\times$ are used interchangeably. Consequently every $m \in M$ is expressed *uniquely* as

$$m = n_1 + n_2 + \cdots + n_k, \qquad n_i \in N_i.$$

6 Free Modules

Definition 6.0: Free module on a set

An R-module F is *free on the subset* $A \subseteq F$ if for every nonzero $x \in F$ there exist *unique* nonzero scalars $r_1, \ldots, r_n \in R$ and unique distinct elements $a_1, \ldots, a_n \in A$ such that

$$x = r_1 a_1 + r_2 a_2 + \cdots + r_n a_n.$$

The set A is called a *basis* or a *set of free generators* of F.

Remark 6.1

1. Notice the definition of a basis A for a module M generated by A in 3.3 is equivalent to saying that A is a set of free generators of M, since by definition of free above we assert that the scalars are unique and they exist for every non-zero element in M.
2. If R is a commutative ring, all bases of any free $R-$module F have the same cardinality.
3. If N is a finitely generated submodule, then it is not necessarily free due to existence of torsion elements. However if M is a free module over a PID, then any submodule is free.

A basis of a free module plays the same dual role as a basis in linear algebra:

- **Spanning:** every element of F is an R-linear combination of basis elements.
- **Linear independence** (uniqueness): $r_1a_1 + \cdots + r_na_n = 0 \;\Rightarrow\; r_1 = \cdots = r_n = 0$.

Example 6.2: Vector spaces are free modules

Every vector space V over a field F is a free F-module: every $x \in V$ is written *uniquely* as a linear combination of the basis vectors, which is exactly the definition of a basis in linear algebra. Moreover, in vector spaces no proper subset of any basis is a spanning set of V, unlike modules that don't force this condition on their bases.

Theorem 6.3: Existence and structure of $F(A)$

For any finite set $A = \{a_1, a_2, \ldots, a_n\}$, there exists a free R-module on A:

$$F(A) \;=\; Ra_1 + Ra_2 + \cdots + Ra_n,$$

in which every element $m \in F(A)$ is uniquely expressed as $m = r_1a_1 + r_2a_2 + \cdots + r_na_n$, $r_i \in R$. Moreover,

$$F(A) \;=\; Ra_1 \oplus Ra_2 \oplus \cdots \oplus Ra_n \;\cong\; R \times R \times \cdots \times R \;=\; R^n.$$

The isomorphism $F(A) \cong R^n$ follows from:

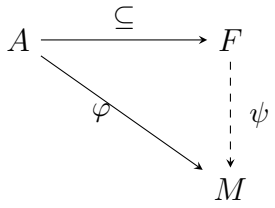
(i) Uniqueness of representation of elements in $F(A)$ makes $F(A)$ an *internal* direct sum $Ra_1 \oplus \cdots \oplus Ra_n$.

(ii) Since $R \cong Ra_i$ as R-modules (via $r \mapsto ra_i$), the internal direct sum is isomorphic to $R \times R \times \cdots \times R = R^n$.

(iii) Note that the theorem implies any two free $R-$modules on the same set are isomorphic since both are isomorphic to R^n.

Theorem 6.4: Universal property of free modules

Let F be a free R-module on A. For any R-module M and any set map $\varphi : A \to M$, there exists a *unique* R-module homomorphism $\psi : F \to M$ extending ϕ (i.e. $\psi|_A = \phi$):



Proof. Let $\phi : A \to M$ be any function.

Consider a function

$$\Psi : F \to M$$

extending ϕ, where F is a free $R-$module and M is any R-modules.

Since F is free on A, every element $b \in F$ can be uniquely expressed as

$$b = a_1 r_1 + a_2 r_2 + \cdots + a_n r_n$$

for some unique $r_i \in R$.

Define Ψ on elements of F by

$$\Psi(b) = \Psi(a_1 r_1 + a_2 r_2 + \cdots + a_n r_n) = \phi(a_1) r_1 + \phi(a_2) r_2 + \cdots + \phi(a_n) r_n.$$

Equivalently,

$$\Psi(b) = \Psi(a_1) r_1 + \Psi(a_2) r_2 + \cdots + \Psi(a_n) r_n.$$

This is clearly an R-module homomorphism.

Moreover, by uniqueness of the coefficients r_i, the map Ψ is uniquely determined, so b cannot be mapped to any other element while still extending ϕ. That is once we know the values of an R-module homomorphism on A its values on every element of the free R-module are uniquely determined, so Ψ is the unique extension of ϕ to all of F. □

Corollary 6.5: Uniqueness of free modules

Any two free R-modules on the same set A are isomorphic and share the universal property.

Note 6.6

Practical consequence of the universal property: to define an R-module homomorphism out of a free module F, it suffices to specify its values on the basis A and extend by linearity. The universal property guarantees the extension is unique. Concretely:

$$\varphi : A \longrightarrow M \qquad \rightsquigarrow \qquad \psi : F \longrightarrow M \quad \text{(unique } R\text{-module hom extending } \varphi).$$

Theorem 6.7: R-linear dependence on free modules

Let R be an integral domain, and let M be a free R-module of finite rank n, then any $n+1$ elements of M are R linearly dependent, that is, for any, $m_1, m_2, \cdots, m_{n+1} \in M$, there exists $r_1, r_2, ..., r_{n+1} \in R$, not all zero, such that

$$r_1 m_1 + r_2 m_2 + \cdots + r_{n+1} m_{n+1} = 0$$

Proof. Since R is an integral domain, The map $\pi\colon R \to F$, $r \mapsto r/1$, is an injective ring homomorphism, so R embeds into the field F, so F can be seen as an R-module. Now, since M is a free R-module of rank n, there is an R-module isomorphism $\varphi : M \to R^n$. In addition, we can embed R^n into F^n the same way we did with R and F, so, F^n can be viewed as an R-module

$$\Psi : R^n \to F^n, \quad (r_1, \ldots, r_n) \mapsto (r_1/1, \ldots, r_n/1)$$

is an injective R-module homomorphism. Composing with φ gives an injective R-module map $\psi : M \to F^n$, so we may regard M as a submodule of F^n.

F^n is not only an $R-$module, but also, a vector space over the field F with $\dim_F F^n = n$. Any collection of more than n vectors in an n-dimensional F-vector space is F-linearly dependent.

Let $m_1, \ldots, m_{n+1} \in M$. Viewing these as elements of F^n (via the embedding ψ), there exists $\alpha_1, \ldots, \alpha_{n+1} \in F$, *not all zero*, such that

$$\alpha_1 m_1 + \alpha_2 m_2 + \cdots + \alpha_{n+1} m_{n+1} = 0 \quad \text{in } F^n.$$

Write $\alpha_i = a_i/b_i$ with $a_i, b_i \in R$ and $b_i \neq 0$. Set

$$D = b_1 b_2 \cdots b_{n+1} \in R, \qquad r_i = a_i \prod_{j \neq i} b_j \in R,$$

so that $D\alpha_i = r_i$ for each i. Multiplying the F-linear relation through by D gives

$$r_1 m_1 + r_2 m_2 + \cdots + r_{n+1} m_{n+1} = 0,$$

with each $r_i \in R$.

Since the α_i are not all zero, fix an index k with $\alpha_k \neq 0$, i.e. $a_k \neq 0$. Then

$$r_k = a_k \prod_{j \neq k} b_j.$$

Each $b_j \neq 0$ and $a_k \neq 0$. Because R is an integral domain it has no zero divisors, so a finite product of nonzero elements is nonzero; hence $r_k \neq 0$, and the r_i are not all zero.

Since for any arbitrary $n+1$ elements, $m_1, m_2, .., m_{n+1}$, we have shown that there must exist, $r_1, r_2, ..., r_{n+1}$, such that □

After introducing R-linear indpendence, we define the *rank* of a free module which is equivalent to the basis in case of when R is an integral domain

Definition 6.8: Rank of any R-module

For any integral domain R the *rank* of a free R-module M is the maximum number of R-linearly independent elements of M.

Note 6.9

If M a vector space is an $F-$module, where F is a field, we refer to the rank of V as the dimension of V defined as the number of elements in any basis of V

6.1 The Matrix of a Linear Transformation

Definition 6.10: Matrix of a linear Transformation

Let $\varphi : V \to W$ be a linear map between free F-modules (vector spaces) with ordered bases $\mathcal{B} = \{v_1, \ldots, v_n\}$ for V and $\mathcal{E} = \{w_1, \ldots, w_m\}$ for W. The *matrix of* φ *with respect to* $\mathcal{B}$ *and* $\mathcal{E}$ is the $m \times n$ matrix $M_{\mathcal{B}}^{\mathcal{E}}(\varphi) = (\alpha_{ij})$ defined by

$$\varphi(v_j) \;=\; \sum_{i=1}^{m} \alpha_{ij}\, w_i,$$

so the j-th column consists of the $\mathcal{E}$-coordinates of $\varphi(v_j)$, that is the scalars in W in the unique linear combination representing $\varphi(v_j)$ with respect to the basis $\mathcal{E}$.

$$M_{\mathcal{B}}^{\mathcal{E}}(\varphi) = \begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{m1} & \alpha_{m2} & \cdots & \alpha_{mn} \end{pmatrix} \longleftarrow j\text{-th column} = \text{coords of } \varphi(v_j)$$

Recovering the map. For any $v = \sum_{i=1}^{n} a_i v_i \in V$, the matrix equation

$$M_{\mathcal{B}}^{\mathcal{E}}(\varphi) \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$$

gives the coordinates of $\varphi(v) = \sum_{i=1}^{m} b_i w_i$ in $\mathcal{E}$.

7 Vector spaces as $F[x]$ modules

In this section we consider a vector space V over F and show how we can make V an $F[x]$-module through linear transformations.

Theorem 7.0

Let V be a vector space over a field F, with

$$S = \{v_1, v_2, v_3, \ldots, v_n\}$$

as a basis. Then

$$V \cong F^n.$$

Proof. Let $\phi : V \to F^n$ be a function where

$$v = a_1 v_1 + a_2 v_2 + \cdots + a_n v_n,$$

given by

$$\phi(v) = (a_1, a_2, a_3, \ldots, a_n).$$

We will show that ϕ is an isomorphism. Since S is a basis, then all linear combinations are vectors in V. By closure of V, hence every scalar tuple

$$(a_1, a_2, \ldots, a_n) \in F^n$$

has an image. Injectivity follows from uniqueness, and

$$\begin{aligned}\phi(v+u) &= (a_1 + b_1, \ldots, a_n + b_n) \\ &= (a_1, \ldots, a_n) + (b_1, \ldots, b_n).\end{aligned}$$

□

So, we can view V as F^n. Now to construct V an $F[x]$-module, we define the action of the indeterminate x on elements of V as a function $x : V \to V$ where

$$x(v+u) = xv + xu,$$

$$x(cu) = c(xu), \qquad c \in F.$$

The action of x satisfies the conditions of a linear transformation T from V to V. So,

$$T(v+u) = T(v) + T(u),$$

$$T(cu) = c(T(u)), \qquad c \in F.$$

We also define

$$T^0 = I, \qquad T^1 = T, \qquad T^2 = T \circ T, \qquad \ldots, \; T^n = T \circ T \circ \cdots \circ T$$

So we can define the action of any polynomial $p(x)$ of degree n in $F[x]$ as follows:

$$\begin{aligned} p(x)v &= a_1 x(v) + a_2 x^2(v) + \cdots + a_n x^n(v) \\ &= a_1 T(v) + a_2 T^2(v) + \cdots + a_n T^n(v) \end{aligned}$$

By our construction above one can see that every pair (V, T) of a vector space and a linear transformation correspond to a vector space as an $F[x]$-module.

We will see next that the structure of an R-module is determined by the structure of ideals of R. There is a special type of integral domain whose ideal structure is known to us, which are principal ideal domains. Recall:

$$F[x] \text{ is a PID.}$$

8 Modules over PIDs

First, we will introduce some concepts relating to Modules that will be of help to introduce the properties of modules over field, which will be of great help in proving Structure theorem, which a corollary of is the fundamental theorem of finite abelian groups.

Definition 8.0: Noetherian R-Module

let R be a ring and let M be a left-R module, M is said to be a Noetherian R-module, or is said to satisfy ACC on submodules, whenever there is an increasing chain of submodules of M

$$M_1 \subseteq M_2 \subseteq M_3 \cdots$$

it stabilizes, i.e, there exists m such that $M_k = M_m$, for all $k \geq m$. In other words, there exists no infinite increasing chain of submodules on M.

We can observe that this gives us a nice result when R is Noetherian, to see this result we need first to see that taking the module to be the ring itself for any ring R, that is, $M = R$ over R, and letting the scalar multiplication to be the normal multiplication of the ring, then we could see that N is a submodule if and only if it is an ideal.

Proof. $\implies$ if N is a submodule it needs to be closed under scalar multiplication, which would mean absorbing the ring elements, and it needs to be an abelian subgroup of the additive group of R, which would enforce it to be an ideal.
$\impliedby$ Clearly any ideal is a submodule, for it is a subgroup of an abelian group, which is the additive group of R, and is closed under multiplicaiton by any ring element (scalar multiplicaiton), and is a subsset of $M = R$, so it is a submodule. □

Now, if R is Noetherian, we will always have an R-module M that is a Noetherian R-module. This is always true by taking $M = R$, and defining the scalar multiplication to be the normal multiplication of the ring R, M will be Noetherian as all its submodules are ideals, and since it is a Noetherian ring, then it follows that M is a Noetherian R-module.

Theorem 8.1: Properties of Noetherian modules

Let R be a ring, and let M be a left R-module then TFAE

1. M is a Noetherian R-module.
2. Any non-empty set of submodules have a maximal element under set inclusion.
3. Every submodule of M is finitely generated.

Proof. $1 \implies 2$ Assume M is Noetherian; then M satisfies the Ascending Chain Condition (ACC) for submodules. Now, let Ψ be a non-empty collection of submodules, if Ψ is finite it must have a maximal element as Ψ is partially ordered under set inclusion, now, assume Ψ is not finite, and assume for the sake of contradiction that Ψ does not have a maximal element.

By the Axiom of Choice, we can construct the following strictly increasing infinite chain of submodules indexed by the natural numbers

$$N_1 \subsetneq N_2 \subsetneq N_3 \subsetneq \dots$$

We can do this because we know that for any N_i, we must have N_k, such that $N_i \subsetneq N_k$ for $i, k \in \mathbb{N}$, otherwise, it would be a maximal element which is not true by assumption, this chain does not stabilize (it is strictly increasing), which is a contradiction as M satisfies ACC for submodules.

$2 \implies 3$ Assume 2, now, let N be a submodule, and Ψ be the collection of all finitely generated submodules of N, this is non-empty as $\{0\} \in \Psi$, now by (2), there exists a maximal element $K \in \Psi$, assume for contradiction that $K \neq N$, let x be an element in $N - K$, since K is finitely generated, then, the submodule L generated by K and x is finitely generated as well, so $L \in \Psi$, and, $K \subsetneq L$, which contradicts that K is the maximum element, so we can't have a maximal element of the set of finitely generated submodules of N that is not N, so, N is finitely generated.

$3 \implies 1$ Let $M_1 \subseteq M_2 \subseteq M_3 \subseteq \dots$ be an ascending chain of submodules of M. Let $N = \bigcup_{i=1}^{\infty} M_i$. We show N is a submodule:

- **Additive Closure:** Let $x, y \in N$. Then $x \in M_j$ and $y \in M_k$ for some indices j, k. Without loss of generality, assume $j \leq k$. Since the submodules form a chain, $M_j \subseteq M_k$, so $x, y \in M_k$. Since M_k is a submodule, $x + y \in M_k \subseteq N$.

- **Scalar Multiplication:** Let $r \in R$ and $x \in N$. Then $x \in M_i$ for some i. Since M_i is

a submodule, $rx \in M_i \subseteq N$.

Thus, N is a submodule of M.

Now assume every submodule of M is finitely generated. Let $M_1 \subseteq M_2 \subseteq M_3 \subseteq \ldots$ be an ascending chain of submodules.

Let $N = \bigcup_{i=1}^{\infty} M_i$. By our assumption, N is finitely generated, say by the set $\{a_1, a_2, \ldots, a_n\}$.

Each generator a_i belongs to N, so by the definition of a union, each $a_i \in M_{j_i}$ for some index j_i. Since there are only finitely many generators, we can define:

$$m = \max\{j_1, j_2, \ldots, j_n\}$$

Because the submodules are nested, $M_{j_i} \subseteq M_m$ for all $i = 1, \ldots, n$. Therefore, all generators $\{a_1, \ldots, a_n\}$ are contained in M_m.

It follows that the submodule they generate, N, is contained in M_m. Since we already know $M_m \subseteq N$, we have:

$$M_m = N$$

For any $k \geq m$, we have $M_m \subseteq M_k \subseteq N$. Since $M_m = N$, it must be that $M_k = M_m$ for all $k \geq m$. Thus, the chain stabilizes, proving M is Noetherian. □

Now, taking the R-module to be R itself with normal multiplication we get the following nice corollary.

Corollary 8.2: Ideals on PIDs

Every non-empty set of ideals on a PID has a maximal element.

This follows clearly since every PID is a Noetherian ring, and by taking the R-module to be the ring itself with the normal multiplication then it follows that it's a Noetherian R-module, and ideals are moudles, so the properties of modules applies here on ideals.

Theorem 8.3

Let R be a Principal Ideal Domain, let M be a free R-module of finite rank n and let N be a submodule of M. Then

(1) N is free of rank m, $m \leq n$ and

(2) there exists a basis $y_1, y_2, \ldots, y_n$ of M so that $a_1y_1, a_2y_2, \ldots, a_my_m$ is a basis of N where $a_1, a_2, \ldots, a_m$ are nonzero elements of R with the divisibility relations

$$a_1 \mid a_2 \mid \cdots \mid a_m.$$

Proof. The theorem is trivial when $N = \{0\}$: it is free of rank 0 and there is nothing to prove. So assume $N \neq \{0\}$.

Step 1: Constructing the Key Ideal and Maximal Element

For each R-module homomorphism $\varphi \in \mathrm{Hom}_R(M, R)$, the image $\varphi(N)$ is a submodule of R, hence an ideal. Since R is a PID every ideal is principal, so $\varphi(N) = \langle a_\varphi \rangle$ for some $a_\varphi \in R$. Define the collection of all such principal ideals:

$$\Sigma = \{\langle a_\varphi \rangle \mid \varphi \in \mathrm{Hom}_R(M, R)\}.$$

Σ is nonempty: The zero homomorphism gives $\langle 0 \rangle \in \Sigma$. Moreover since M is free with basis $\{x_1, \ldots, x_n\}$, the coordinate projections $\pi_i(m) = r_i$ (where $m = \sum r_j x_j$) are nontrivial homomorphisms. Since $N \neq \{0\}$, there exists some i with $\pi_i(N) \neq \langle 0 \rangle$, so Σ contains more than just the trivial ideal.

Σ has a maximal element: Σ is a nonempty collection of ideals in a PID. A PID is a Noetherian ring, meaning every ascending chain of ideals stabilizes. Therefore Σ has a maximal element, that is, there exists $\nu \in \mathrm{Hom}_R(M, R)$ such that $\nu(N) = \langle a_1 \rangle$ is not properly contained in any other element of Σ.

$a_1 \neq 0$: Since $\langle a_1 \rangle$ is maximal in Σ and Σ contains ideals strictly larger than $\langle 0 \rangle$ (from the

projection maps above), we must have $\langle a_1 \rangle \neq \langle 0 \rangle$, so $a_1 \neq 0$.

Step 2: Finding the Element y_1

Since $\nu(N) = \langle a_1 \rangle$, there exists $y \in N$ such that $\nu(y) = a_1$.

a_1 divides $\varphi(y)$ for every $\varphi \in \mathbf{Hom}_R(M, R)$: Let $\varphi \in \mathrm{Hom}_R(M, R)$ be arbitrary and let d be a generator of the principal ideal $\langle a_1, \varphi(y) \rangle$ in R. Then:

$$d \mid a_1, \quad d \mid \varphi(y), \qquad \text{and} \qquad d = r_1 a_1 + r_2 \varphi(y) \text{ for some } r_1, r_2 \in R.$$

Define the homomorphism $\psi = r_1 \nu + r_2 \varphi : M \to R$. Then:

$$\psi(y) = r_1 \nu(y) + r_2 \varphi(y) = r_1 a_1 + r_2 \varphi(y) = d.$$

So $d \in \psi(N)$, hence $\langle d \rangle \subseteq \psi(N)$. But $d \mid a_1$ means $\langle a_1 \rangle \subseteq \langle d \rangle$, so:

$$\langle a_1 \rangle \subseteq \langle d \rangle \subseteq \psi(N).$$

By maximality of $\langle a_1 \rangle$ in Σ we must have $\langle a_1 \rangle = \langle d \rangle = \psi(N)$. In particular $\langle a_1 \rangle = \langle d \rangle$ means $d \mid \varphi(y)$ and $a_1 \mid d$, together giving $a_1 \mid \varphi(y)$.

Constructing y_1: Applying the above to the coordinate projections, write $\pi_i(y) = a_1 b_i$ for some $b_i \in R$, $1 \leq i \leq n$. Define:

$$y_1 = \sum_{i=1}^{n} b_i x_i \in M.$$

Claim: $a_1 y_1 = y$. Compute:

$$a_1 y_1 = a_1 \sum_{i=1}^{n} b_i x_i = \sum_{i=1}^{n} (a_1 b_i) x_i = \sum_{i=1}^{n} \pi_i(y) x_i = y.$$

The last equality uses the fact that $y = \sum_i \pi_i(y) x_i$ by definition of the coordinate projections.

Claim: $\nu(y_1) = 1$. Since $a_1 = \nu(y) = \nu(a_1 y_1) = a_1 \nu(y_1)$ and $a_1 \neq 0$ in the integral domain

R, we can cancel to get $\nu(y_1) = 1$.

Step 3: Splitting M and N

We claim:

$$\textbf{(a)} \quad M = Ry_1 \oplus \ker \nu,$$
$$\textbf{(b)} \quad N = Ra_1 y_1 \oplus (N \cap \ker \nu).$$

Proof of (a): Take any $x \in M$ and write:

$$x = \nu(x)y_1 + (x - \nu(x)y_1).$$

The first summand is in Ry_1. For the second, compute:

$$\nu(x - \nu(x)y_1) = \nu(x) - \nu(x)\nu(y_1) = \nu(x) - \nu(x) \cdot 1 = 0,$$

so $x - \nu(x)y_1 \in \ker \nu$. This shows $M = Ry_1 + \ker \nu$. For directness: suppose $ry_1 \in \ker \nu$ for some $r \in R$. Then $0 = \nu(ry_1) = r\nu(y_1) = r$, so $r = 0$. Hence $Ry_1 \cap \ker \nu = \{0\}$ and the sum is direct.

Proof of (b): Take any $x' \in N$. Since $\nu(x') \in \nu(N) = \langle a_1 \rangle$, write $\nu(x') = ba_1$ for some $b \in R$. Decompose:

$$x' = ba_1 y_1 + (x' - ba_1 y_1).$$

The first summand is in $Ra_1 y_1$. For the second: $x' - ba_1 y_1 \in \ker \nu$ by the same calculation as in (a), and it lies in N since $ba_1 y_1 = by \in N$ and $x' \in N$. So $x' - ba_1 y_1 \in N \cap \ker \nu$. Directness follows from directness of (a) since $Ra_1 y_1 \subseteq Ry_1$.

Step 4: Proving Part (1) N is Free of Rank $m \leq n$

We induct on the rank m of N.

Base case $m = 0$**:** N is a torsion module. But $N \subseteq M$ and M is free hence torsion free, so $N = \{0\}$, which is free of rank 0.

Inductive step $m > 0$**:** From the direct sum decomposition (b):

$$N = Ra_1y_1 \oplus (N \cap \ker \nu).$$

Since $a_1 \neq 0$ and R is an integral domain, $Ra_1y_1 \cong R$ is free of rank 1. The submodule $N \cap \ker \nu$ has rank $m - 1$. Moreover $\ker \nu$ is free of rank $n - 1$ by decomposition (a). By the induction hypothesis applied to $N \cap \ker \nu \subseteq \ker \nu$, the module $N \cap \ker \nu$ is free of rank $m - 1$. Therefore:

$$N = Ra_1y_1 \oplus (N \cap \ker \nu)$$

is free of rank $1 + (m - 1) = m$, and $m \leq n$ since $N \subseteq M$.

Step 5: Proving Part (2) The Divisibility Chain

We induct on n, the rank of M.

Base case $n = 0$**:** $M = \{0\}$, so $N = \{0\}$ and there is nothing to prove.

Inductive step: By decomposition (a), $\ker \nu$ is free of rank $n - 1$. Apply the induction hypothesis to $\ker \nu$ (playing the role of M) and its submodule $\ker \nu \cap N$ (playing the role of N). We obtain:

- a basis $y_2, y_3, \ldots, y_n$ of $\ker \nu$,
- elements $a_2, a_3, \ldots, a_m \in R$ such that $a_2y_2, a_3y_3, \ldots, a_my_m$ is a basis for $N \cap \ker \nu$,
- the divisibility chain $a_2 \mid a_3 \mid \cdots \mid a_m$.

Since the sums in (a) and (b) are direct, $\{y_1, y_2, \ldots, y_n\}$ is a basis for M and $\{a_1y_1, a_2y_2, \ldots, a_my_m\}$ is a basis for N.

It remains to show $a_1 \mid a_2$: Define a homomorphism $\varphi : M \to R$ by:

$$\varphi(y_1) = 1, \quad \varphi(y_2) = 1, \quad \varphi(y_i) = 0 \text{ for all } i > 2.$$

This is well-defined since $\{y_i\}$ is a basis. Then:

$$a_1 = \nu(a_1 y_1) \in \nu(N) = \langle a_1 \rangle \subseteq \varphi(N),$$

so $\langle a_1 \rangle \subseteq \varphi(N)$. By maximality of $\langle a_1 \rangle$ in Σ we get $\langle a_1 \rangle = \varphi(N)$. Now:

$$a_2 = \varphi(a_2 y_2) \in \varphi(N) = \langle a_1 \rangle,$$

so $a_2 \in \langle a_1 \rangle$, which means $a_1 \mid a_2$. Combined with $a_2 \mid a_3 \mid \cdots \mid a_m$ from the induction hypothesis, we obtain the full chain:

$$a_1 \mid a_2 \mid a_3 \mid \cdots \mid a_m. \qquad \square$$

9 Structure Theorem for finitely generated modules over PID

In this section, we aim to prove the structure theorem for modules over rings whose ideals structure is simple and known to us which are PIDs. This theorem reflects how the ideals structure of a ring R determines the structure of the R-module. Proving the structure theorem for modules over the PID $\mathbb{Z}$ yields what so called the fundamental theorem of finite abelian groups as a corollary. Moreover, we shall show in the next section that by proving the structure theorem for modules over the PID $F[x]$ we obtain important results on matrices canonical form in particular: Rational canonical forms. We can motivate the following discussion by looking at any cyclic module M over a ring R, for this we can define a module homomorphism $\pi : R \to M$, as $\pi(a) = ax$, such that $a \in R$, and $M = Rx$, as it is cyclic, we will see that

$$M \cong R/\ker \pi$$

, by first isomorphism theorem, if M is torsion-free, it is easy to see that $\ker \pi = \{0\}$, which means $M \cong R$, which is a free module of rank 1. In the follwing discussion we will extend this to finitely generated modules over PIDs.

Theorem 9.0: Fundamental Theorem of Finitely Generated Modules over a PID Invariant Factor Form

Let R be a principal ideal domain and let M be a finitely generated R-module.

(1) M is isomorphic to a finite direct sum of cyclic R-modules. More precisely, there exist an integer $r \geq 0$ and nonzero non-unit elements $a_1, a_2, \ldots, a_m \in R$ satisfying the divisibility chain

$$a_1 \mid a_2 \mid \cdots \mid a_m,$$

such that

$$M \cong R^r \oplus R/\langle a_1 \rangle \oplus R/\langle a_2 \rangle \oplus \cdots \oplus R/\langle a_m \rangle.$$

(2) M is torsion-free if and only if M is a free R-module.

(3) The torsion submodule of M is

$$\mathrm{Tor}(M) \cong R/\langle a_1 \rangle \oplus R/\langle a_2 \rangle \oplus \cdots \oplus R/\langle a_m \rangle.$$

In particular, M is a torsion module if and only if $r = 0$, in which case the annihator of M is the principal ideal $\langle a_m \rangle$.

Proof. We proceed in several steps.

Step 1: Constructing a surjection from a free module onto M.

Since M is finitely generated, let $\{x_1, x_2, \ldots, x_n\}$ be a generating set of M of minimal cardinality n. Let $F = R^n$ be the free R-module of rank n with standard basis $\{b_1, b_2, \ldots, b_n\}$. Define the R-module homomorphism

$$\pi : F \longrightarrow M, \qquad \pi(b_i) = x_i \quad \text{for all } i = 1, \ldots, n,$$

extended R-linearly to all of F. Since $x_1, \ldots, x_n$ generate M, the map π is surjective. By

the First Isomorphism Theorem for R-modules,

$$M \cong F/\ker\pi = R^n/\ker\pi.$$

Step 2: Applying the structure theorem for submodules of free modules over a PID.

The kernel $\ker\pi$ is a submodule of the free module $F = R^n$. Since R is a PID, every submodule of a finitely generated free R-module is itself free by Theorem 8.3. Moreover, by that same theorem applied to $F = R^n$ and the submodule $\ker\pi$, there exists a basis $\{y_1, y_2, \ldots, y_n\}$ of R^n and elements $a_1, a_2, \ldots, a_m \in R$ (with $m \leq n$) satisfying

$$a_1 \mid a_2 \mid \cdots \mid a_m,$$

such that $\{a_1y_1, a_2y_2, \ldots, a_my_m\}$ is a basis for $\ker\pi$. Concretely, this means:

$$\ker\pi = Ra_1y_1 \oplus Ra_2y_2 \oplus \cdots \oplus Ra_my_m.$$

Hence we may write the quotient as

$$M \;\cong\; R^n/\ker\pi \;=\; \frac{Ry_1 \oplus Ry_2 \oplus \cdots \oplus Ry_n}{Ra_1y_1 \oplus Ra_2y_2 \oplus \cdots \oplus Ra_my_m}. \tag{1}$$

Step 3: Identifying the quotient via a natural surjection.

To evaluate the right-hand side of (1), we construct an explicit surjective R-module homomorphism

$$\varphi : Ry_1 \oplus Ry_2 \oplus \cdots \oplus Ry_n \longrightarrow R/\langle a_1\rangle \oplus R/\langle a_2\rangle \oplus \cdots \oplus R/\langle a_m\rangle \oplus R^{n-m}$$

defined, for $(\alpha_1 y_1, \alpha_2 y_2, \ldots, \alpha_n y_n)$ with $\alpha_i \in R$, by

$$\varphi(\alpha_1 y_1, \ldots, \alpha_n y_n) \;=\; \Big(\alpha_1 \bmod a_1 \;\ldots,\; \alpha_m \bmod a_m,\; \alpha_{m+1},\; \ldots,\; \alpha_n\Big).$$

Surjectivity of φ. Let $(\bar{c}_1, \ldots, \bar{c}_m, d_{m+1}, \ldots, d_n)$ be an arbitrary element of the target, where $\bar{c}_i \in R/\langle a_i \rangle$ and $d_j \in R$. Pick any representative $c_i \in R$ for each $\bar{c}_i$, such that $c_i \bmod a_i = \bar{c}_i$, Then $\varphi(c_1 y_1, \ldots, c_m y_m, d_{m+1} y_{m+1}, \ldots, d_n y_n)$ equals the given element, so φ is surjective.

Kernel of φ. An element $(\alpha_1 y_1, \ldots, \alpha_n y_n)$ lies in $\ker \varphi$ if and only if

$$\alpha_i \bmod a_i = 0 \;\text{ in } R/\langle a_i \rangle \quad \text{for } i = 1, \ldots, m, \qquad \text{and} \qquad \alpha_j = 0 \quad \text{for } j = m+1, \ldots, n.$$

The first condition is equivalent to $a_i \mid \alpha_i$, i.e. $\alpha_i \in \langle a_i \rangle$, for each $i = 1, \ldots, m$. Writing $\alpha_i = a_i r_i$ for some $r_i \in R$, we see that

$$\ker \varphi \;=\; Ra_1 y_1 \oplus Ra_2 y_2 \oplus \cdots \oplus Ra_m y_m \oplus \{0\} \oplus \cdots \oplus \{0\}.$$

This is precisely the submodule $Ra_1 y_1 \oplus \cdots \oplus Ra_m y_m$, which equals $\ker \pi$ by Step 2.

By the First Isomorphism Theorem applied to φ, we obtain

$$\frac{Ry_1 \oplus \cdots \oplus Ry_n}{Ra_1 y_1 \oplus \cdots \oplus Ra_m y_m} \;\cong\; R/\langle a_1 \rangle \oplus \cdots \oplus R/\langle a_m \rangle \oplus R^{n-m}.$$

Combining with (1):

$$M \;\cong\; R/\langle a_1 \rangle \oplus R/\langle a_2 \rangle \oplus \cdots \oplus R/\langle a_m \rangle \oplus R^{n-m}. \tag{2}$$

Step 4: Removing unit factors and setting $r = n - m$.

If any a_i is a unit in R, then $\langle a_i \rangle = R$, so $R/\langle a_i \rangle = 0$, and that summand contributes nothing

to the decomposition. We may therefore discard all such trivial summands. After removing the unit a_i's, the remaining non-unit elements (which we relabel $a_1, \ldots, a_m$ by absorbing the discarded indices) still satisfy $a_1 \mid a_2 \mid \cdots \mid a_m$. Setting $r = n - m$ yields the decomposition in part (1):

$$M \;\cong\; R^r \oplus R/\langle a_1\rangle \oplus \cdots \oplus R/\langle a_m\rangle.$$

Step 5: Proof of part (2).

Recall that M is *torsion-free* if the only element annihilated by a nonzero ring element is $0 \in M$.

Any nonzero non-unit $a \in R$, the quotient module $R/\langle a\rangle$ is a torsion R-module, because a annihilates every element. Hence if M is torsion-free, it cannot contain any summand of the form $R/\langle a_i\rangle$, so, by the decomposition of part (1) we will have that

$$M \cong R^n$$

which is free, so for modules over PIDs any finitely generated torsion free module is free.

Conversely, the free module R^r is torsion-free: if $r \cdot (c_1, \ldots, c_r) = \mathbf{0}$ for some nonzero $r \in R$, then $rc_i = 0$ in R for all i, which forces $c_i = 0$ since R is an integral domain (all PIDs are domains), so if M is free, it is isomorphic to R^n, and since R^n is torsion free, then, M is torsion free. Therefore M is torsion-free if and only if $m = 0$, i.e. $M \cong R^r$, proving (2).

Step 6: Proof of part (3).

We identify $\mathrm{Tor}(M)$ within the decomposition (2). As argued in Step 5, each summand $R/\langle a_i\rangle$ (for $i = 1, \ldots, m$) is entirely torsion, while R^r is entirely torsion-free. Since the direct sum of torsion modules is torsion, and the torsion submodule of a direct sum $A \oplus B$

is $\mathrm{Tor}(A) \oplus \mathrm{Tor}(B)$ (you can check that), we conclude:

$$\mathrm{Tor}(M) \;\cong\; R/\langle a_1\rangle \oplus R/\langle a_2\rangle \oplus \cdots \oplus R/\langle a_m\rangle.$$

In particular, M is a torsion module (i.e. $\mathrm{Tor}(M) = M$) if and only if $r = 0$.

It remains to show that when $r = 0$, the annihilator of M is the ideal $\langle a_m\rangle$. Recall that $\mathrm{Ann}(M) = \{s \in R : s \cdot x = 0 \text{ for all } x \in M\}$. Since $a_1 \mid a_2 \mid \cdots \mid a_m$, we have $\langle a_m\rangle \subseteq \langle a_{m-1}\rangle \subseteq \cdots \subseteq \langle a_1\rangle$ as ideals. For any $s \in \mathrm{Ann}(M)$, s must annihilate the summand $R/\langle a_m\rangle$ in particular, which means $s \in \langle a_m\rangle$ (since $\mathrm{Ann}(R/\langle a\rangle) = \langle a\rangle$ for any nonzero $a \in R$). Conversely, since a_m annihilates $R/\langle a_i\rangle$ for every $i (because a_i \mid a_m$ implies $a_m \in \langle a_i\rangle$, so $a_m \cdot \bar{x} = \overline{a_m x} = \bar{0}$ in $R/\langle a_i\rangle)$, we get $a_m \in \mathrm{Ann}(M)$, i.e. $\langle a_m\rangle \subseteq \mathrm{Ann}(M)$. Therefore $\mathrm{Ann}(M) = \langle a_m\rangle$. □

Definition 9.1: Free Rank and Invariant Factors

The integer r is called the *free rank* or the *Betti number* of M and the elements $a_1, a_2, \ldots, a_m \in R$ (defined up to multiplication by units in R) are called the *invariant factors* of M.

Theorem 9.2: Fundamental Theorem, Existence: Elementary Divisor Form

Let R be a P.I.D. and let M be a finitely generated R-module. Then M is the direct sum of a finite number of cyclic modules whose annihilators are either (0) or generated by powers of primes in R, i.e.,

$$M \cong R^r \oplus R/(p_1^{\alpha_1}) \oplus R/(p_2^{\alpha_2}) \oplus \cdots \oplus R/(p_t^{\alpha_t})$$

where $r \geq 0$ is an integer and $p_1^{\alpha_1}, \ldots, p_t^{\alpha_t}$ are positive powers of (not necessarily distinct) primes in R.

Definition 9.3: Elementary Divisors

Let R be a P.I.D. and let M be a finitely generated R-module as in theorem 9.2. The prime powers $p_1^{\alpha_1}, \ldots, p_t^{\alpha_t}$ (defined up to multiplication by units in R) are called the *elementary divisors* of M

Theorem 9.4: Fundamental Theorem: Uniqueness

Let R be a PID

(1) Two finitely generated R-modules M_1 and M_2 are isomorphic if and only if they have the same free rank and the same list of invariant factors.

(2) Two finitely generated R-modules M_1 and M_2 are isomorphic if and only if they have the same free rank and the same list of elementary divisors.

Corollary 9.5: Fundamental Theorem of Finite Abelian groups

Every finite Abelian group is a direct product of cyclic groups of prime-power order. Moreover, the number of terms in the product and the orders of the cyclic groups are uniquely determined by the group.

Proof. Since $\mathbb{Z}$ is a PID, taking $R = \mathbb{Z}$ and applying theorems 9.0, 9.2, and 9.3, by taking any finite abelian group as a $\mathbb{Z}$ module, any module taken this way will be finitely generated over a PID, therefore, the theorem holds, as the direct product of groups is just the direct sum of modules over $\mathbb{Z}$. □

10 Rational Canonical Form

The structure of an $F[x]-$module (its cyclic decomposition) which we obtained by a vector space V over a field F associated with a linear transformation T which represents that action of the polynomials in $F[x]$ helps us to know a lot of information about the linear transformation T (rational canonical form). Hence this is an instance of when the structure of a space being acted upon in this case V helps revealing information about the algebraic object acting on it in this case T. This is exactly similar to the case of Representation theory, where we use vector spaces (space being acted upon via groups) to obtain information about the algebraic objects acting on the space (the groups), which helps us easily understand the structure of groups using an easy structure which is vector spaces.

Note 10.0

In this section, we treat a linear transformation T of any sapce V in a very similar way to its matrix A with respect to a fixed basis according to the definition stated in 6.1

First, we begin with the following lemma:

Lemma 10.1

Let V be a vector space over a field F. Let $T : V \to V$ be a linear transformation. The following are equivalent:

1. λ is an eigenvalue of T.
2. $(\lambda I - T)$ is a singular linear transformation of V.
3. $\det(\lambda I - T) = 0$

Proof. (1 $\iff$ 2) From linear algebra, we know that λ is an eigenvalue of T iff λ is a solution of the equation $Tv = xv$ iff $Tv = \lambda v$ iff $(\lambda I - T)v = 0$ iff the corresponding non-zero eigenvector v is in the kernel of the transformation $(\lambda I - T)$, yielding that $(\lambda I - T)$ is singular that is the kernel of the transformation contains non-zero vectors.

(2 $\iff$ 3) A transformation has determinant zero iff it is non-invertible iff the nullity(T) is non-zero iff the kernel contains non-zero vectors iff $(\lambda I - T)$ is singular. □

Definition 10.2: Characterstic polynomial

Let V be vector space of dim n over a field F with a fixed basis $\mathcal{B}$. Let $T : V \to V$ be a linear transformation. Let x be indeterminate over F. The polynomial $\det(xI - T)$ is called the characterstic polynomial of T denoted by $c_T(x)$.

Moreover, if A is the matrix of T with respect to the basis $\mathcal{B}$ we call the polynomial $\det(xI - A)$ the characterstic polynomial of A denoted by $c_A(x)$.

Lemma 10.3: Degree of characterstic polynomial

Let F be a field. Let V a vector space be an $F[x]-$module of dim n with a fixed basis $\mathcal{B}$. Let $T : V \to V$ be a linear transformation with matrix A with respect to $\mathcal{B}$. Then

$$\deg(c_A(x)) = \deg(c_T(x)) = n.$$

Lemma 10.4

Any non-zero free $F[x]-$module V is an infinite dimensional vector space over F.

Proof. As V is a free $F[x]-$module, let

$$\mathcal{E} = \{e_1, e_2, \ldots, e_n\}$$

be a basis for V over $F[x]$.Then any $v \in V$ is of the form

$$v = P_1(x)e_1 + P_2(x)e_2 + \cdots + P_n(x)e_n.$$

It is clear that

$$S = \{1, x, x^2, \ldots\}$$

is an infinite basis for $F[x]$ over F. Hence each polynomial $P_i(x)$ can be written as

$$P_i(x) = a_0 + a_1x + \cdots + a_mx^m.$$

So, every polynomial can be written using finitely many x^i elements from the basis S.

However, no finite basis in $F[x]$ can represent all polynomials. Since $F[x]$ cannot be expressed using a finite basis over F, neither can V, because every element in V is already expressed as a linear combination of polynomials from $F[x]$. Therefore V is infinite dimensional over F. □

By the converse of the previous lemma we get that if an $F[x]-$module V is a finite dimensional vector space then it must be finitely generated and non-free. Hence, by the structure theorem we know that V must contain some torsion elements. In the following lemma we prove even a stronger statement.

Lemma 10.5

An $F[x]-$module V with a finite dimension as a vector space over F is a torsion module

Proof. By the structure theorem

$$V \cong F[x]^r \oplus \bigoplus_{i=1}^{m} F[x]/I_i$$

Suppose $r \neq 0$. We will show that $V \cong F[x]^r$. Consider the map

$$\phi : F[x]^r \to F[x]^r \oplus \bigoplus_{i=1}^{m} F[x]/I_i$$

It is clear that ϕ is injective. Since $F[x]^r \oplus \bigoplus_{i=1}^{m} F[x]/I_i \cong V$ via a map ψ. Then the composition

$$\psi \circ \phi : F[x]^r \to V$$

is an injective map. Since It follows that there is a subspace of V isomorphic to $F[x]^r$. Since $F[x]$ is infinite dimensional vector space over F, then V is also infinite dimensional over F contradicting our assumoptions. Therefore $r = 0$, and V is a torsion module. □

Let V be a finite dimensional vector space with a linear transformation T. By the previous

lemma we get that the annihilator of V as an $F[x]-$module is the ideal I_m. Since $F[x]$ is a PID, we have a unique monic polynomial $m(x)$ of minimal degree generating $\text{Ann}(M)$ such that $m(T)$ is the zero linear transformation. Now, we define the minimal polynomial of a linear transformation.

Definition 10.6: The minimal polynomial

Let F be a field. Let V a vector space be an $F[x]-$module of dim n with a fixed basis $\mathcal{B}$. The unique monic polynomial generating the annihilator of V is called the unique minimal polynomial of T denoted by $m_T(x)$. The unique monic polynomial of smallest degree which when evaluated at the matrix A of T is the zero matrix is called the minimal polynomial of A denoted by $m_A(x)$.

Note 10.7

Note that while constructing the rational canonical form, we don't assume that the eigenvalues of a linear transformation are necessarily in the field we are working in. This leads to the study of characterstic and minimal polynomials in extensions of the field F to obtain information about their splitting fields and roots. Or assuming that F is algebraically closed. In this paper we are only considering in the roots (which may not exist) in the field of interest.

Theorem 10.8: Cayley-Hamilton Theorem

Let F be a field. Let V a vector space be an $F[x]-$module of dim n with a linear transformation T. Then

$$m_T(x) \mid c_T(x).$$

The Cayley-Hmailton theorem yields that the degree of the minimal polynomial of T is at most the dimension of the space n. We will prove it the theorem shortly after we introduce the rational canonical form. Before that, we introduce the definition of a direct sum of matrices.

Definition 10.9: Direct sum of matrices

The direct sum of two matrices $A_{m\times n}$ and $B_{r\times s}$ denoted by $A \bigoplus B$ is a matrix

$$M = \begin{pmatrix} A & \mathbf{0} \\ \mathbf{0} & B \end{pmatrix}$$

of size $(m+s)\times(n+r)$ such that the top left corner is the matrix A and the bottom right corner is the matrix B, with all other entries zeroes.

We will see later that the rational canonical form of a matrix is simply a direct sum of matrices.

Definition 10.10: Similar matrices

Let A and B be any two matrices over a field F. We say A and B are similar iff there exists an invertible matrix P such that $A = P^{-1}BA$. We say A is similar to B.

Now, we are ready to define the rational canonical form of matrices of the linear transformation T associated with the vector space V as an $F[x]-$module using the structure theorem.

Basis of the summands $F[x]/\langle a_i(x)\rangle$

First we have this isomorphism:

$$V \cong \bigoplus_{i=1}^{m} F[x]/\langle a_i(x)\rangle \tag{3}$$

Let $a_i(x) = b_0 + b_1x + b_2x^2 + \cdots + b_kx^k$, where $b_i \in F$. By the division algorithm in $F[x]$ we get that every polynomial in the vector space $F[x]$ over F is equivalent to the remainder $r(x)$ upon dividing by $a_i(x)$ in $F[x]/\langle a_i(x)\rangle$. Hence the elements of $F[x]/\langle a_i(x)\rangle$ are exactly the remainders $r(x)$ where $\deg(r(x)) = 0, 1, 2, \ldots, k-1$. Therefore the set $\{1, \bar{x}, \bar{x}^2, \ldots, \bar{x}^k\}$ is a basis for each summand $F[x]/\langle a_i(x)\rangle$, where $\bar{x} = x + \langle a_i(x)\rangle$. For simplicity we refer to $a_i(x)$ by $a(x)$

Action of x

By 10, it is clear that every $F[x]/\langle a_i(x)\rangle$ could be also seen as an $F[x]-$module. Hence, we define the action of x on the basis elements $\{1, \bar{x}, \bar{x}^2, \ldots, \bar{x}^k\}$ in each summand as follows:

$$\begin{aligned} 1 &\mapsto \bar{x} \\ \bar{x} &\mapsto \bar{x}^2 \\ &\vdots \\ \bar{x}^{k-1} &\mapsto \bar{x}^k \end{aligned}$$

where the last term $\bar{x}^k = -b_0 - b_1 x - \cdots - b_{k-1}x^{k-1}$. To see this, we compute

$$\begin{aligned} a(\bar{x}) &= b_0 + \langle a(x)\rangle + b_1 x + \langle a(x)\rangle + \cdots + b_k x^k \langle a(x)\rangle \\ &= (b_0 + b_1 x + \cdots + b_k x^k) + \langle a(x)\rangle \\ &= 0 + \langle a(x)\rangle \end{aligned}$$

Thus $a(\bar{x}) = 0$ in $F[x]/\langle a(x)\rangle$. Now, since multiplication by x is a linear transformation in $F[x]/\langle a(x)\rangle$, we compute its associated matrix with respect to the basis $\{1, \bar{x}, \bar{x}^2, \ldots, \bar{x}^k\}$ to be in this form:

$$\begin{pmatrix} 0 & 0 & \cdots & \cdots & 0 & -b_0 \\ 1 & 0 & \cdots & \cdots & 0 & -b_1 \\ 0 & 1 & \ddots & & \vdots & -b_2 \\ \vdots & \ddots & \ddots & \ddots & 0 & \vdots \\ 0 & \cdots & 0 & 1 & 0 & -b_{k-2} \\ 0 & \cdots & \cdots & 0 & 1 & -b_{k-1} \end{pmatrix}$$

For any monic polynomial $p(x) \in F[x]$ we have such a matrix that we call *the companion matrix of the polynomial* $a_i(x)$.

Definition 10.11

Let $a(x) = b_0 + b_1x + b_2x^2 + \cdots + b_kx^k$ be any monic polynomial over a field F. The **companion matrix** of the polynomial $a(x)$ is a $k \times k$ matrix with $1'$s down the main diagonal, $-b_0, -b_1, \ldots, -b_{k-1}$ in the last column and zeroes elsewhere.

The matrix of the linear transformation T

First we state the following lemma

Lemma 10.12:

Let V be a vector space, where V is a direct sum of vector spaces V_i.

$$V = \bigoplus_{i=1}^{m} V_i$$

where every V_i has $\mathcal{B}_i$ as a basis. The union $\mathcal{B} = \bigcup_{i=1}^{m} \mathcal{B}_i$ is a basis for V.

The linear transformation T is defined by the action of x on every element in V. By 10, every basis element of the set $S = \{1, \bar{x}, \bar{x}^2, \ldots, \bar{x}^k\}$ is mapped to a vector in V.

$$(\bar{x}, 0, 0, 0, 0, \underbrace{\ldots}_{m \text{ times}}, 0) \mapsto v_i$$

We denote $(\bar{x}, 0, 0, 0, 0, \underbrace{\ldots}_{m \text{ times}}, 0)$ by $\bar{x}$ for simplicity.

We collect all v_i for each element in the basis S of the summand $F[x]/\langle a_i(x)\rangle$ to form the linearly independent set $B_i \subseteq V$. Note that V is the direct sum of the span of every $\mathcal{B}_i$, that is $V = \bigoplus_{i=1}^{m} V_i$ because $V \cong \bigoplus_{i=1}^{m} F[x]/\langle a_i(x)$. Now, we define the action of x on the elements of B_i, . By 10, the image of the action of x on $\bar{x}$ is the action of T on the image of $\bar{x}$, that is $v_i \in B_i$. Therefore, T acts on v_i as the companion matrix of $a_i(x)$ since this is how we defined the action of x on $\bar{x}$. Note that by our construction above, the basis elements of each summand is mapped to B_i, then the basis of the direct sum is mapped to

the union of B_i. By 10, the union of B_i for each $a_i(x)$ is a basis for V, because the sum on the right of the isomorphism is direct, then the basis of the direct sum is the union of the bases of the summands. Moreover, The elements of the basis of the direct sum are mapped to the basis elements of V by the isomorphism. Now, with respect to the basis B, the linear transformation T has as matrix the direct sum of the companion matrices for the invariant factors that is:

$$M = \begin{pmatrix} C_{a_1(x)} & 0 & \cdots & 0 \\ 0 & C_{a_2(x)} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & C_{a_m(x)} \end{pmatrix}$$

Note that by the uniqueness of the invariant factors $a_i(x)$ of V, we get that the matrix M is also unique for a linear transformation T.

Definition 10.13: Rational Canonical Form

A matrix A is said to be in the rational canonical form if it is the direct sum of companion matrices of monic polynomials $a_1(x), a_2(x), \ldots, a_m(x)$ of degree at least 1, with

$$a_1 \mid a_2 \mid \cdots \mid a_m$$

The polynomials $a_i(x)$ are called the invariant factors of the rational canonical form. Such a matrix is also called a block diagonal matrix with the diagonal blocks being the companion matrices of $a_i(x)$. Moreover, A rational canonical form for a linear transformation T is a matrix representing T which is in rational canonical form.

Definition 10.14: Invariant factors of a matrix

The invariant factors of an $n \times n$ matrix A are the invariant factors of its rational canonical form.

This construction yields the following two results:

Theorem 10.15: Rational Canonical Form for Linear Transformations

Let V be a finite dimensional vector space over the field F and let T be a linear transformation of V.

1. There is a basis for V with respect to which the matrix for T is in rational canonical form, i.e., is a block diagonal matrix whose diagonal blocks are the companion matrices for monic polynomials

$$a_1(x), a_2(x), \ldots, a_m(x)$$

of degree at least one with

$$a_1(x) \mid a_2(x) \mid \cdots \mid a_m(x).$$

2. The rational canonical form for T is unique.

Theorem 10.16: Rational Canonical Form for Matrices

Let A be an $n \times n$ matrix over the field F.

1. The matrix A is similar to a matrix in rational canonical form, i.e., there is an invertible $n \times n$ matrix P over F such that

$$P^{-1}AP$$

is a block diagonal matrix whose diagonal blocks are the companion matrices for monic polynomials

$$a_1(x), a_2(x), \ldots, a_m(x)$$

of degree at least one with

$$a_1(x) \mid a_2(x) \mid \cdots \mid a_m(x).$$

2. The rational canonical form for A is unique.

The construction we illustrated above proves the existence of a basis for V with respect to which a linear transformation T has a simple matrix. So, we now know that every linear transformation T has a rational canonical form. It remains to show the uniqueness part.

Proof. Let V be a finite dimensional vector space over the field F and let T be a linear transformation of V . By the structure theorem

$$V \cong \bigoplus_{i=1}^{m} F[x]/\langle a_i(x)\rangle$$

Suppose there are monic polynomials $b_1(x), b_2(x), \ldots, b_t(x) \in F[x]$ of degree at least one with $b_i \mid b_{i+1}$. Suppose that for some basis $\mathcal{E}$ of V, the matrix M of the linear transformation T with respect to the basis $\mathcal{E}$ is the direct sum of the companion matrices of $b_i(x)$.

$$[T]_{\mathcal{E}} = \begin{pmatrix} C_{b_1(x)} & & & \\ & C_{b_2(x)} & & \\ & & \ddots & \\ & & & C_{b_t(x)} \end{pmatrix}$$

where every $C_{b_i}(x)$

$$C_{b_i(x)} = \begin{pmatrix} 0 & 0 & 0 & \cdots & 0 & -c_0 \\ 1 & 0 & 0 & \cdots & 0 & -c_1 \\ 0 & 1 & 0 & \cdots & 0 & -c_2 \\ 0 & 0 & 1 & \cdots & 0 & -c_3 \\ \vdots & & & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & -c_{d_{i-1}} \end{pmatrix}$$

Hence the columns of M corresponds to the images of the basis vectors of $\mathcal{E}$. Let

$$\mathcal{E} = \{e_{11}, e_{12}, \ldots, e_{1d_1}, e_{21}, e_{22}, \ldots, e_{2d_2}, \ldots, e_{t1}, \ldots e_{td_t}\}$$

be a basis of V. We partition $\mathcal{E}$ into t many subsets $\mathcal{E}_i$ of d_i-many elements where $\mathcal{E} = \bigcup_{i=1}^{t} \mathcal{E}_i$. Since every $\mathcal{E}_i$ is a linearly independent set of d_i elements, then it is a basis for the subspace $D_i = \text{Span}(\mathcal{E}_i)$. Therefore, by construction of M it is easy to see that every companion matrix $C_{b_i(x)}$'s columns represent the images of the basis $\mathcal{E}_i$ of the subspace D_i under T, that is every companion matrix represent the action of T on D_i. Now, we will show that D_i is a cyclic $F[x]-$module. Pick the first vector e_{i1} in the basis of D_i, the image of e_{i1} under T is represented by the first column of $C_{b_i}(x)$, which has 1 in the second entry. So,

$$\begin{aligned} T(e_{i1}) &= e_{i2} \\ T(e_{i2}) &= e_{i3} \\ &\vdots \\ T(e_{id_{i-1}}) &= e_{id_i} \\ T(e_{id_i}) &= -c_0\, e_{i1} - c_1\, e_{i2} - \cdots - c_{d_{i-1}}\, e_{id_i} \end{aligned}$$

Hence, every basis element could be obtained by repeatedly applying the transformation T to the first element j_1, which is the same as multiplying by the polynomial x. Thus every element in D_i is of the form $a_1 e_{i1} + a_2 T(e_{e_{i1}}) + a_3 T^2(e_{i1}) + \cdots + a_{d_{i-1}} T^{d_{i-1}}(e_{i1})$, where every $a_j \in F$. Note that this proves D_i are T stable subspaces i.e. $T(D_i) \subseteq D_i$ and that every D_i is a cyclic $F[x]$-module generated by the first basis element e_{i1}, and $b_i(x)$ is the annihilator of D_i. Since $T^k(e_{i1}) = e_{ik+1}$ for $0 \leq k < d_i$. and

$$T^{d_i}(e_{i1}) = T(e_{id_i}) = -c_0 e_{i1} - c_1 e_{i2} - \cdots - c_{d_{i-1}} e_{id_i}$$

So,

$$\begin{aligned} b_i(T)(e_{i1}) &= T^{d_i}(e_{i1}) + c_{d_i-1}T^{d_i-1}(e_{i1}) + \cdots + c_1T(e_{i1}) + c_0e_{i1} \\ &= \underbrace{(-c_0e_{i1} - c_1e_{i2} - \cdots - c_{d_i-1}e_{id_i})}_{T^{d_i}(e_{i1})} + c_{d_i-1}e_{id_i} + \cdots + c_1e_{i2} + c_0e_{i1} \\ &= (-c_0 + c_0)e_{i1} + (-c_1 + c_1)e_{i2} + \cdots + (-c_{d_i-1} + c_{d_i-1})e_{id_i} = 0. \end{aligned}$$

So, V is a direct sum of cyclic modules in two different ways. Thus, by uniqueness of the invariant factors, we get that $a_i(x)$ is the same as $b_i(x)$ up to units. Since we assumed that they are monic polynomials then $a_i(x) = b_i(x)$.

□

We then obtain the following corollary:

Corollary 10.17

Let $W = \langle w \rangle$ be a cyclic submodule of the $F[x]$-module V with the linear transformation T. Let $b(x)$ be the annihilator of W with degree n.

Then the set

$$\{T^{n-1}(w), T^{n-2}(w), \ldots, T(w), w\}$$

is a basis for W.

Theorem 10.18

Let S and T be linear transformations of an $n-$dimensional vector space V. Then the following are equivalent:

1. S and T are similar linear transformations.
2. The $F[x]$-modules obtained from V via S and via T are isomorphic $F[x]$-modules.
3. S and T have the same rational canonical form.

Proof. (1) $\Rightarrow$ (2) Assume there is a nonsingular linear transformation U such that

$$S = UTU^{-1}.$$

Thus, $T = U^{-1}SU$ Consider the following linear transformation from V to itself

$$U : V \to V$$

is also an $F[x]$-module homomorphism, where x acts on the first V via T and on the second via S. It remains to show that the action of T on any $v \in V$ is equal to the action of S on $U(x)$. We proceed as follows:

$$U(xv) = U(Tv) = UT(v) = U(U^{-1}(S(U(v)))) = S(U(v)) = x(U(v)).$$

Hence this is an $F[x]$-module isomorphism of the two modules in (2).

(2) $\Rightarrow$ (3) Assume (2) holds and denote by V_1 the vector space V made into an $F[x]$-module via S and denote by V_2 the space V made into an $F[x]$-module via T. Since

$$V_1 \cong V_2$$

as $F[x]$-modules they have the same list of invariant factors. Thus S and T have a common rational canonical form.

(3) $\Rightarrow$ (1) Assume (3) holds. Since S and T have the same matrix representation with respect to some choice of (possibly different) bases of V by assumption, they are, up to a change of basis, the same linear transformation of V. To see this let $\mathcal{B}_1 = \{v_1, v_2, \ldots, v_n\}$ and $\mathcal{B}_2 = \{w_1, w_2, \ldots, w_n\}$ be two bases of V. Consider the two linear transformations S with respect to the basis $\mathcal{B}_1$ and T with respect to the basis $\mathcal{B}_2$ with the same rational canonical form A. Therefore by definition of a matrix of a linear transformation in 6.1, we have that for any $v_i \in \mathcal{B}_1$ and any $w_i \in \mathcal{B}_2$:

$$S(v_i) = \sum_{j=1}^{n} a_{ij} v_j \qquad , T(w_i) = \sum_{j=1}^{n} a_{ij} w_j$$

Consider a linear transformation $U : V \to V$ such that $U(v_i) = w_i$. Since $\mathcal{B}_1$ and $\mathcal{B}_2$ are two bases, the U is a well defined linear transformation. We claim that $S = U^{-1}TU$. To prove the claim, we proceed by properties of the linear transformation U as follows :

$$\begin{aligned} U^{-1}TU(v_i) &= U^{-1}(T(w_i)) \\ &= U^{-1}(\sum_{j=1}^{n} a_{ij}w_j) \\ &= \sum_{j=1}^{n} U^{-1}(a_{ij}w_j)) \\ &= \sum_{j=1}^{n} a_{ij}U^{-1}(w_j) \\ &= \sum_{j=1}^{n} a_{ij}v_i = S(v_i) \end{aligned}$$

Hence we proved that $S = U^{-1}TU$, and so S and T are similar. □

Lemma 10.19:

Let $a(x) \in F[x]$ be any monic polynomial.

1. The characteristic polynomial of the companion matrix of $a(x)$ is $a(x)$.
2. If M is the block diagonal matrix

$$M = \begin{pmatrix} A_1 & 0 & \cdots & 0 \\ 0 & A_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & A_k \end{pmatrix},$$

given by the direct sum of matrices

$$A_1, A_2, \ldots, A_k,$$

then the characteristic polynomial of M is the product of the characteristic polynomials of $A_1, A_2, \ldots, A_k$.

Proof. (1) Let $a(x)$ be any monic polynomial in $F[x]$ of degree m. We proceed by induction on m.

Base case: $m = 2$. Consider $a(x) = a_0 + a_x + a_2x^2$. We obtain a 2×2 companion matrix as follows:

$$C_{a(x)} = \begin{pmatrix} 0 & -a_0 \\ 1 & -a_1 \end{pmatrix}.$$

Hence,

$$xI - C_{a(x)} = \begin{pmatrix} x & a_0 \\ -1 & x + a_1 \end{pmatrix}.$$

By definition of determinant we get that

$$c_A(x) = \det(xI - C_{a(x)}) = x(x + a_1) + a_0 = x^2 + a_1x + a_0 = a(x$$

IH: Suppose the statement is true for any monic polynomial of degree m. Consider $a(x)$ a monic polynomial of degree $m + 1$. We obtain the following comapnion matrix

$$C_{a(x)} = \begin{pmatrix} 0 & 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & 0 & \cdots & 0 & -a_2 \\ \vdots & \ddots & \ddots & \ddots & \vdots & \vdots \\ 0 & \cdots & 0 & 1 & 0 & -a_m \end{pmatrix}.$$

So,

$$xI - C_{a(x)} = \begin{pmatrix} x & 0 & 0 & \cdots & 0 & a_0 \\ -1 & x & 0 & \cdots & 0 & a_1 \\ 0 & -1 & x & \cdots & 0 & a_2 \\ \vdots & \ddots & \ddots & \ddots & \vdots & \vdots \\ 0 & \cdots & 0 & -1 & 0 & x + a_m \end{pmatrix}.$$

Using the cofactor expansion of the first row we get that

$$\det(xI - C_{a(x)}) = xC_{11} + a_0 \cdot (-1)^m(\det(M))$$

where M is an upper triangular matrix of -1 on its diagonal. So, $\det(M) = (-1)^m$ and C_{11} is the determinant of the matrix

$$\begin{pmatrix} x & 0 & \cdots & 0 & a_1 \\ -1 & x & \cdots & 0 & a_2 \\ 0 & -1 & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & x & a_{m-1} \\ 0 & \cdots & 0 & -1 & a_m \end{pmatrix}$$

which is excatly $\det(xI - C_{a_1(x)})$, where $a_1(x)$ is a polynomial of degree m. Applying the induction hypothesis we get that $C_{11} = a_1(x) = a_1 + a_2x + \cdots + x^m$. Therefore

$$\begin{aligned} \det(xI - C_{a(x)}) &= xC_{11} + a_0 \cdot (-1)^m(-1)^m \\ &= x(a_1(x)) + a_0 \\ &= a_0 + a_1x + a_2x^2 + \cdots + x^{m+1} = a(x) \end{aligned}$$

as desired. □

Before we prove (2), we state the following fact:

Fact 10.20:

Let M be a diagonal block square matrix. The determinant of M is the product of the determinantes of its blocks. That is if

$$M = \begin{pmatrix} A_{11} & 0 & \cdots & 0 \\ 0 & A_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & A_{mm} \end{pmatrix}$$

Then,

$$\det(M) = \det(A_{11}) \times \det(A_{22}) \times \cdots \times \det(A_{mm})$$

Proof. (2) Let M be a block diagonal matrix given by the direct sum of matrices $A_1, A_2, \ldots, A_k$. By the previous Fact 10, it remains to show that $xI - M$ is a block diagonal matrix. To see this note that $xI - M$ only changes the entries of each block of M, since the diagonal of I is the only non-zero entries. Thus the diagonal elements of M are the only entries that change by subtraction. Hence, $xI - M$ is a block matrix where each block is $xI - A_{ii}$. Therefore,

$$\det(xI - M) = \det(xI - A_{11}) \times \det(xI - A_{22}) \times \cdots \times \det(xI - A_{kk})$$

and so the characterstic polynomial of M is the product of characterstic polynomials of the blocks. □

Theorem 10.21

Let A be an $n \times n$ matrix over the field F.

1. The characteristic polynomial of A is the product of all the invariant factors of A.
2. **(The Cayley–Hamilton Theorem)** The minimal polynomial of A divides the characteristic polynomial of A.
3. The characteristic polynomial of A divides some power of the minimal polynomial of A. In particular these polynomials have the same roots, not counting multiplicities.

Note 10.22

1. The same statements are true if the matrix A is replaced by a linear transformation T of an n-dimensional vector space over F.
2. By definition 10.3 of the characteristic polynomial of a matrix A, the eigenvalues of A are exactly the roots of $c_A(x)$ in F.

Proof. Let V a vector space be an $F[x]-$module with a linear transformation T and a fixed basis B. Let A be the matrix of T with respect to B.

(1) By definition the invariant factors of A are excatly the invariant factors of its rational canonical form. By the structure theroem,

$$V \cong \bigoplus_{i=1}^{m} F[x]/\langle a_i(x)\rangle$$

with $a_i \mid a_{i+1}$, and A is the matrix of the linear transformation T. Then, the rational canonical form of A is the direct sum of the companion matrices of each $a_i(x)$. Since the characterstic polynomial of A is the characterstic polynomial of its rational canonical form, then by lemma 10

$$c_A(x) = \Pi_{i=1}^{m} a_i(x)$$

as desired.

(2) Since the unique monic polynomial generating the annihilator of V is the minimal polynomial of T, then it is the minimal polynomial of A. By the structure theorem the annihilator of V is the last invariant factor in the cyclic decomposition of V which is $a_m(x)$. Since $c_A(x) = \Pi_{i=1}^{m} a_i(x)$ by (1) and $m_A(x) = a_m(x)$. Thus, $m_A(x) \mid c_A(x)$ as desired.

(3) From (2) it is clear that any root of the minimal polynomial of A is also a root of the characteristic polynomial $c_A(x)$. It remains to show the converse. Let λ be any root of $c_A(x)$

in F. By definition of the characteristic polynomial, we get that λ is an eigenvalue of A. So,

$$Av = \lambda v \tag{4}$$

By definition the minimal polynomial annihilates A, so we get that

$$\begin{aligned} & m_A(A)v = m_A(\lambda)v \\ \implies & \mathbf{0}v = m_A(\lambda)v \end{aligned}$$

Since v is non-zero it must be that $m_A(\lambda) = 0$ showing that λ is a root of the minimal polynomial as desired. $\square$

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